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Each of these modes helps control how data is managed in the file, and it is important to choose the appropriate mode depending on whether the file should be modified or preserved.

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3. Error Handling in File Processing

When working with files, errors can occur for various reasons, such as the file not existing, insufficient permissions, or problems while reading or writing data. Proper error handling is essential to make sure that your program behaves correctly even in the event of a file error.

3.1 Checking for Errors When Opening Files

Always check if the file is successfully opened by verifying the return value of `fopen()`. If the file cannot be opened, `fopen()` returns `NULL`, and you should handle this error appropriately.

```
FILE *file = fopen("data.txt", "r");

if (file == NULL) {

    perror("Error opening file");

    exit(1);  // Exit program or take appropriate action

}
```